

ENV 790.30 - Time Series Analysis for Energy Data | Spring 2025

Assignment 3 - Due date 02/03/26

Sadie Kasten

Directions

You should open the .rmd file corresponding to this assignment on RStudio. The file is available on our class repository on Github.

Once you have the file open on your local machine the first thing you will do is rename the file such that it includes your first and last name (e.g., “LuanaLima_TSA_A03_Sp25.Rmd”). Then change “Student Name” on line 4 with your name.

Then you will start working through the assignment by **creating code and output** that answer each question. Be sure to use this assignment document. Your report should contain the answer to each question and any plots/tables you obtained (when applicable).

Please keep this R code chunk options for the report. It is easier for us to grade when we can see code and output together. And the tidy.opts will make sure that line breaks on your code chunks are automatically added for better visualization.

When you have completed the assignment, **Knit** the text and code into a single PDF file. Submit this pdf using Sakai.

Questions

Consider the same data you used for A2 from the spreadsheet “Table_10.1_Renewable_Energy_Production_and_Consumption”. The data comes from the US Energy Information and Administration and corresponds to the December 2025 Monthly Energy Review. This time you will work only with the following columns: **Total Renewable Energy Production**; and **Hydroelectric Power Consumption**.

Create a data frame structure with these two time series only.

R packages needed for this assignment: “forecast”, “tseries”, and “Kendall”. Install these packages, if you haven’t done yet. Do not forget to load them before running your script, since they are NOT default packages.\

```
#Load/install required package here
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method      from
##   as.zoo.data.frame zoo
```

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(forecast)
library(readxl)
library(openxlsx)
library(ggplot2)
library(Kendall)
library(cowplot)

rm(list = ls())
```

```
#creating a dataframe from the A2 data
#Importing data set without change the original file using read.xlsx
energy_data1 <- read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source")
```

```
## New names:
## * '' -> '...1'
## * '' -> '...2'
## * '' -> '...3'
## * '' -> '...4'
## * '' -> '...5'
## * '' -> '...6'
## * '' -> '...7'
## * '' -> '...8'
## * '' -> '...9'
## * '' -> '...10'
## * '' -> '...11'
## * '' -> '...12'
## * '' -> '...13'
## * '' -> '...14'
```

```
#Now let's extract the column names from row 11
read_col_names <- read_excel(path="./Data/Table_10.1_Renewable_Energy_Production_and_Consumption_by_Source", sheet = 11)
```

```
## New names:
## * '' -> '...1'
## * '' -> '...2'
## * '' -> '...3'
## * '' -> '...4'
## * '' -> '...5'
## * '' -> '...6'
## * '' -> '...7'
## * '' -> '...8'
## * '' -> '...9'
## * '' -> '...10'
## * '' -> '...11'
```

```
## * '' -> '...12'
## * '' -> '...13'
## * '' -> '...14'
```

```
colnames(energy_data1) <- read_col_names

#variables I want
selected_vars <- energy_data1[,c("Month", "Total Renewable Energy Production", "Hydroelectric Power Consumption")]

#turn into a dataframe
new_df <- as.data.frame(selected_vars)
head(new_df)
```

```
##           Month Total Renewable Energy Production Hydroelectric Power Consumption
## 1 1973-01-01                219.839                89.562
## 2 1973-02-01                197.330                79.544
## 3 1973-03-01                218.686                88.284
## 4 1973-04-01                209.330                83.152
## 5 1973-05-01                215.982                85.643
## 6 1973-06-01                208.249                82.060
```

```
##Trend Component
```

Q1

For each series (Total Renewable Production and Hydroelectric Consumption) create three plots arranged in a row (side-by-side): (1) time series plot, (2) ACF, (3) PACF. Use `cowplot::plot_grid()` to place them in a grid.

```
#transform into timeseries
ts_A3 <- ts(new_df[,1:2], start=c(1973,1),frequency = 12)
head(ts_A3)
```

```
##           Total Renewable Energy Production Hydroelectric Power Consumption
## Jan 1973                219.839                89.562
## Feb 1973                197.330                79.544
## Mar 1973                218.686                88.284
## Apr 1973                209.330                83.152
## May 1973                215.982                85.643
## Jun 1973                208.249                82.060
```

```
#time series plots
ts1 <- autoplot(ts_A3[,1],
  main = colnames(ts_A3)[1],
  xlab = "Year",
  ylab = "Energy Production")

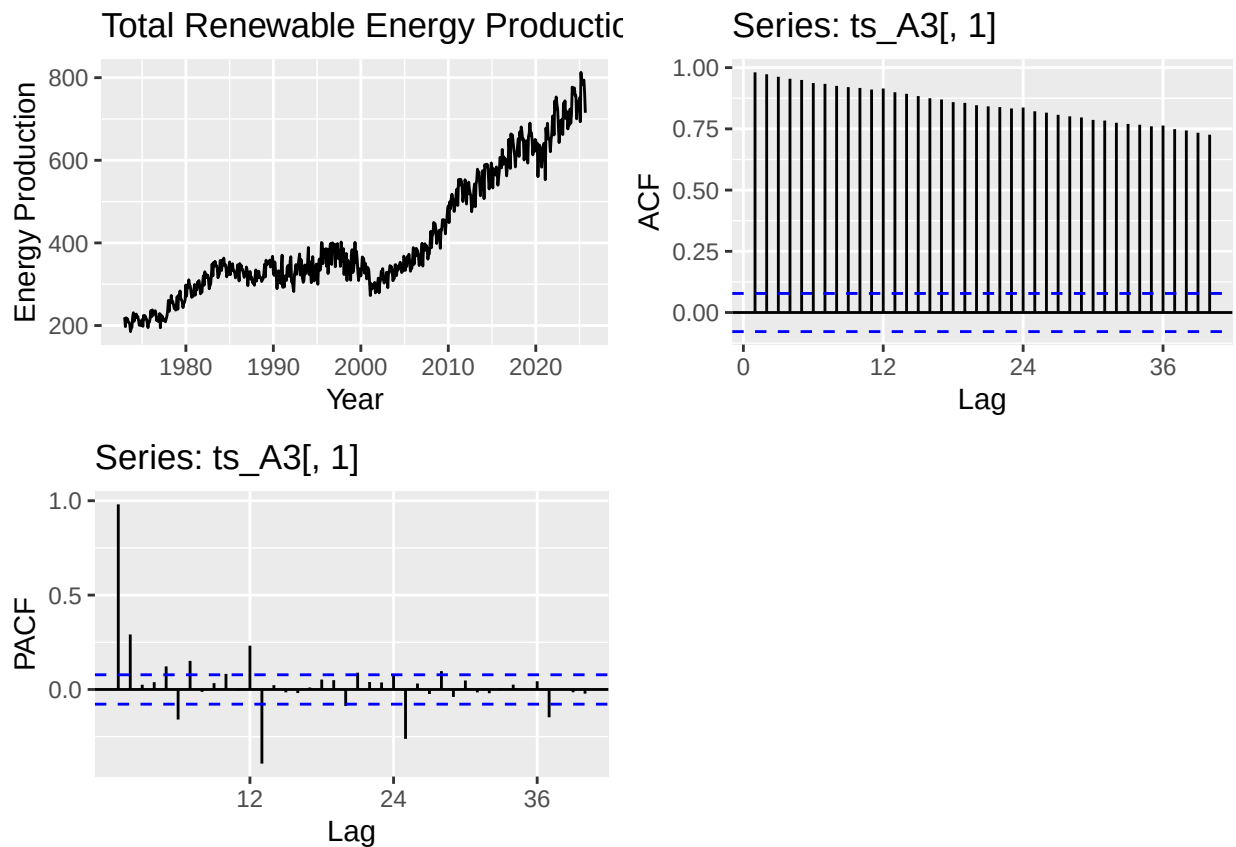
ts2 <- autoplot(ts_A3[,2],
  main = colnames(ts_A3)[2],
  xlab = "Year",
  ylab = "Energy Consumption")

#ACF plots
```

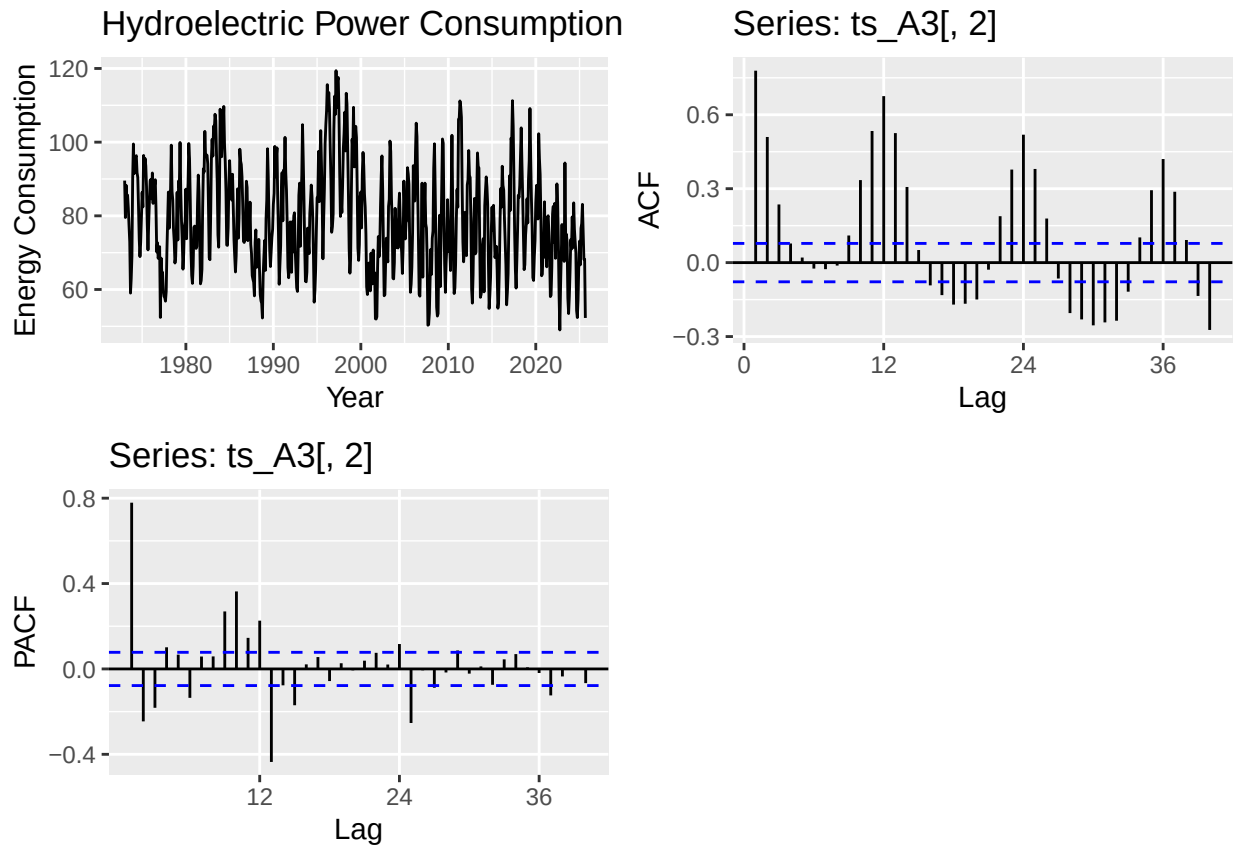
```
ACF1 <- ggAcf(ts_A3[,1],lag.max=40,plot=T)
ACF2 <- ggAcf(ts_A3[,2], lag.max=40, plot=T)

#PACF plots
PACF1 <- ggPacf(ts_A3[,1],lag.max=40,plot=T)
PACF2 <- ggPacf(ts_A3[,2], lag.max=40, plot=T)

#Plots, side by side using cowplot
plot_grid(ts1, ACF1, PACF1)
```



```
plot_grid(ts2, ACF2, PACF2)
```



Q2

From the plot in Q1, do the series Total Renewable Energy Production and Hydroelectric Power Consumption appear to have a trend? If yes, what kind of trend?

Renewable energy production appears to have an increasing trend, which would be a deterministic trend. Linear(ish) increase over time. Hydroelectric power consumption appears to be seasonally influenced which makes determining a trend difficult. The seasonal oscillation seems to stay around some mean so it's possible there is not a trend (doesn't seem to increase or decrease over time).

Q3

Use the `lm()` function to fit a linear trend to the two time series. Ask R to print the summary of the regression. Interpret the regression output, i.e., slope and intercept. Save the regression coefficients for further analysis.

```
nobs <- nrow(energy_data1)
#Create vector t
t<- c(1:nobs)

#Fit trend for Renewable Energy
RE_linear_trend <- lm(new_df[,2]~t)# plus one bc of date column in the data frame
summary(RE_linear_trend)
```

```
##
```

```
## Call:
## lm(formula = new_df[, 2] ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -154.81  -39.55   12.52   41.49  171.15
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 171.44868    5.11085   33.55  <2e-16 ***
## t           0.74999     0.01397   53.69  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 64.22 on 631 degrees of freedom
## Multiple R-squared:  0.8204, Adjusted R-squared:  0.8201
## F-statistic: 2883 on 1 and 631 DF,  p-value: < 2.2e-16
```

```
#store coefficients
RE_beta0 <- as.numeric(RE_linear_trend$coefficients[1]) # intercept
RE_beta1 <- as.numeric(RE_linear_trend$coefficients[2]) # slope
```

Renewable has a high R^2 and low p-value.

```
#Fit trend for Hydropower data
HP_linear_trend <- lm(new_df[,3]~t)# plus one bc of date column in the data frame
summary(HP_linear_trend)
```

```
##
## Call:
## lm(formula = new_df[, 3] ~ t)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.190 -10.214  -0.715    8.909   39.723
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 83.223802    1.110552  74.939  < 2e-16 ***
## t          -0.012199    0.003035  -4.019 6.55e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.95 on 631 degrees of freedom
## Multiple R-squared:  0.02496, Adjusted R-squared:  0.02342
## F-statistic: 16.15 on 1 and 631 DF,  p-value: 6.547e-05
```

```
#store reg coefficient
HP_beta0 <- as.numeric(HP_linear_trend$coefficients[1]) # intercept
HP_beta1 <- as.numeric(HP_linear_trend$coefficients[2]) # slope
```

Hydropower has a low R^2 and high p-value.

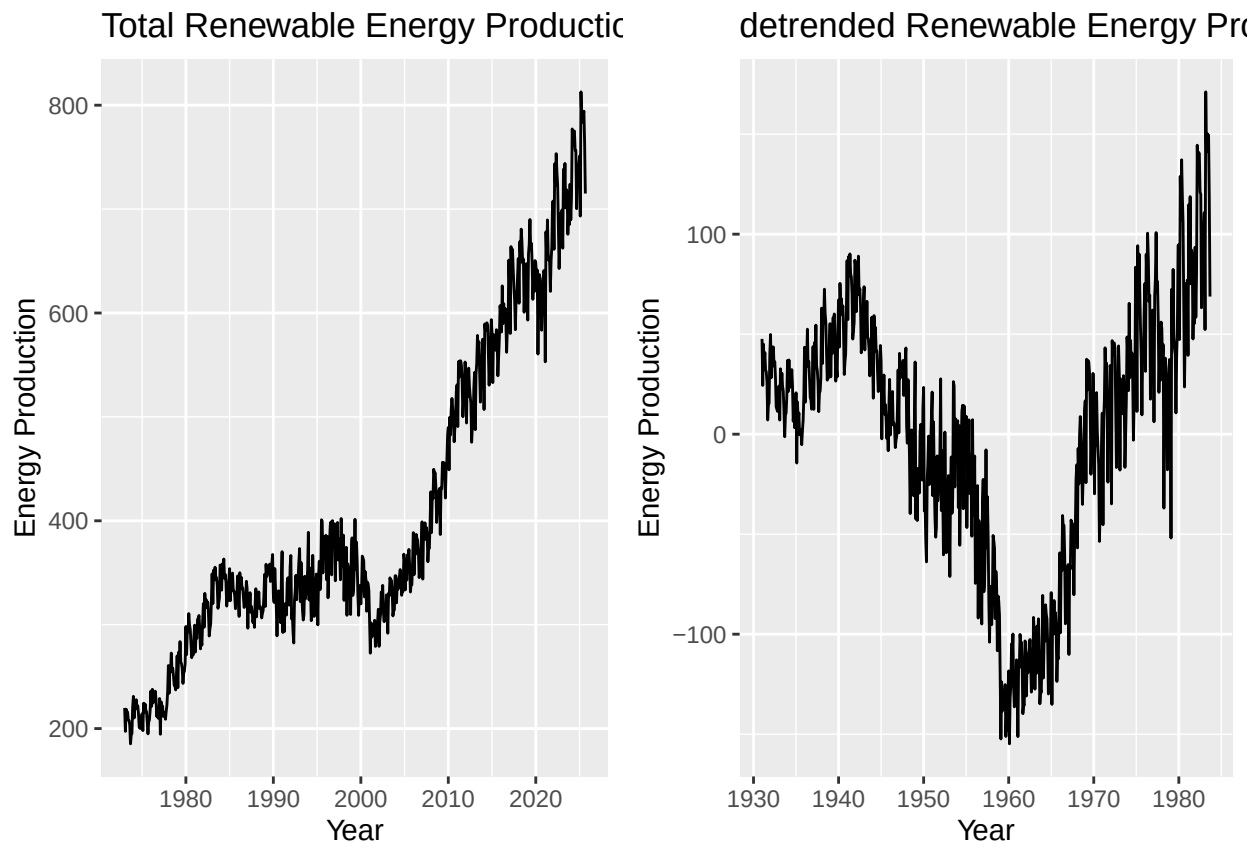
Q4

Use the regression coefficients to detrend each series (subtract fitted linear trend). Plot detrended series and compare with the original time series from Q1. Describe what changed.

```
detrend_RE <- new_df[,2] - (RE_beta0 + RE_beta1*t)
#transform to ts
ts_detrend_RE <- ts(detrend_RE,frequency = 12, start=c(1931,1))
class(ts_detrend_RE)
```

```
## [1] "ts"
```

```
ts3 <- autoplot(ts_detrend_RE,
  main = "detrended Renewable Energy Production",
  xlab = "Year",
  ylab = "Energy Production")
plot_grid(ts1, ts3)
```

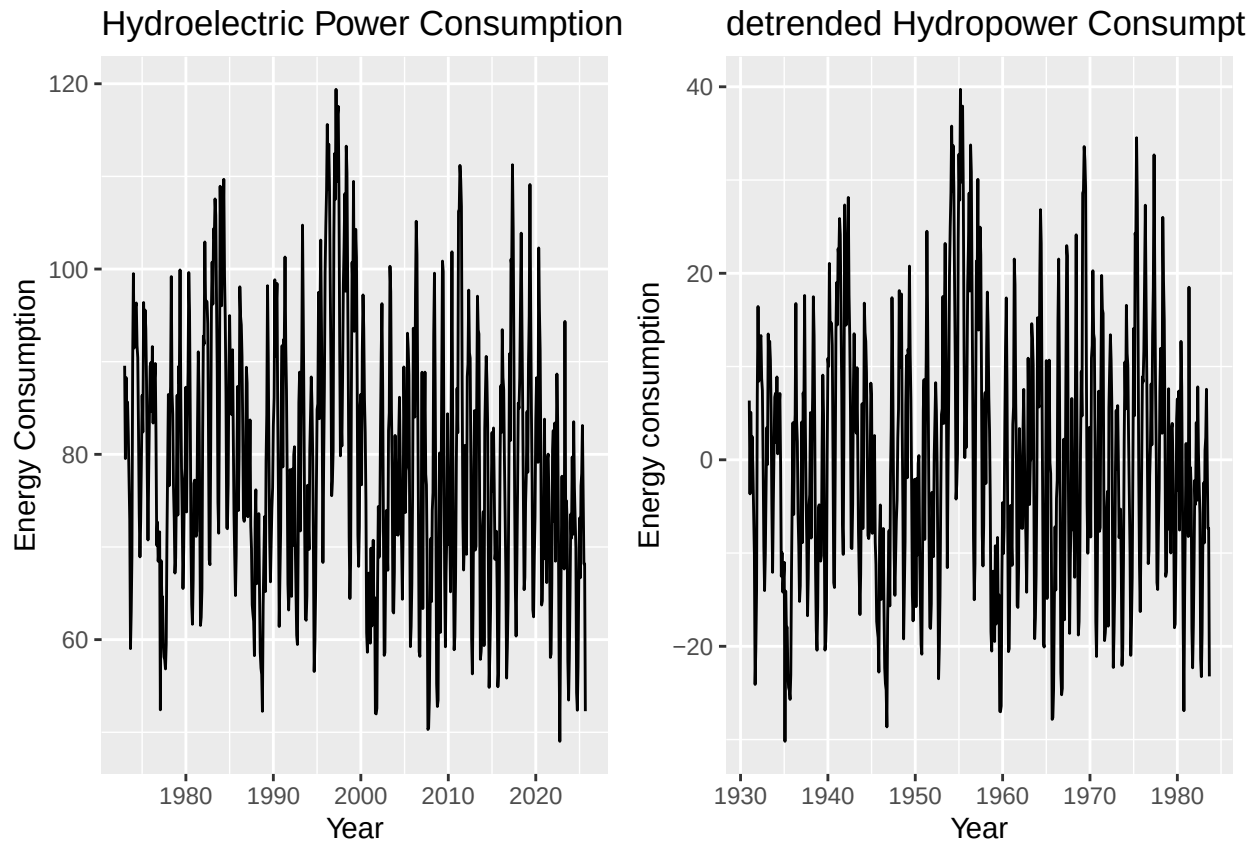


For renewable it mostly removed the linear upwards trend (though it kinda looks still observed). I kind of expected it to be flat but it is not! has a v shape now. Moved the series down to around 0.

```
detrend_HP <- new_df[,3] - (HP_beta0 + HP_beta1*t)
#transform to ts
ts_detrend_HP <- ts(detrend_HP,frequency = 12, start=c(1931,1))
```

```
ts4 <- autoplot(ts_detrend_HP,
  main = "detrended Hydropower Consumption",
  xlab = "Year",
  ylab = "Energy consumption")

plot_grid(ts2, ts4)
```



For hydropower the graph looks the same. It slightly leveled it out (maybe there was a negative trend in more recent years). Now the series is centered at 0.

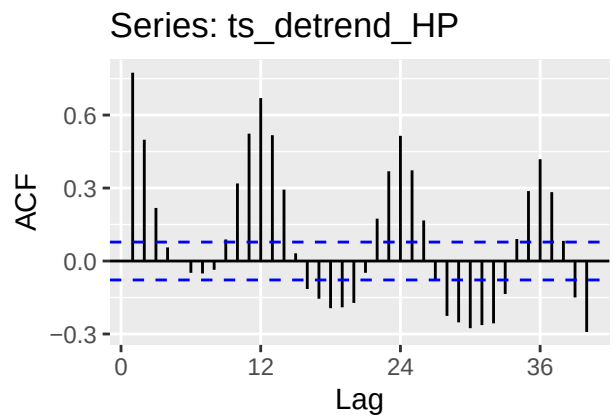
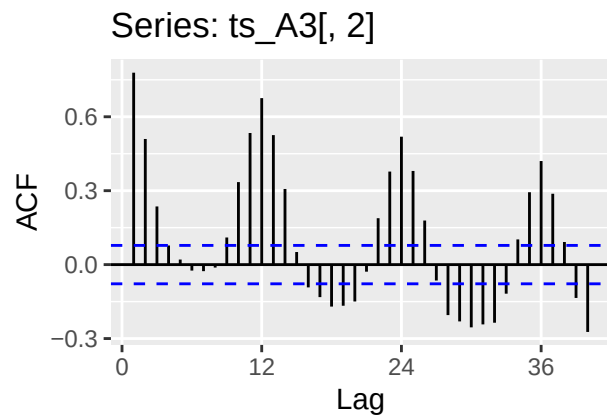
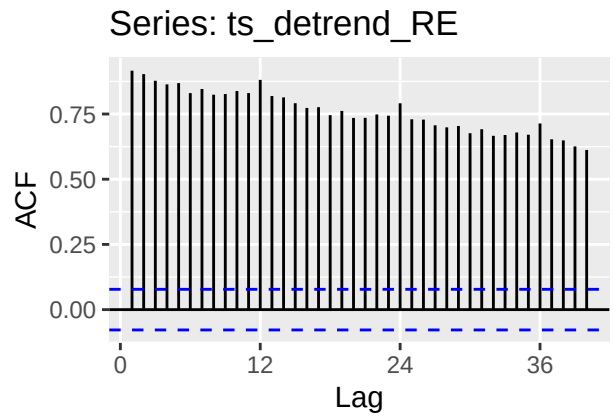
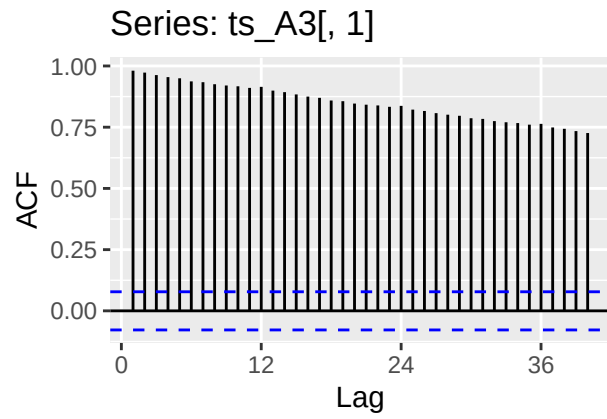
Q5

Plot ACF and PACF for the detrended series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side to make it easier to compare. Did the plots change? How?

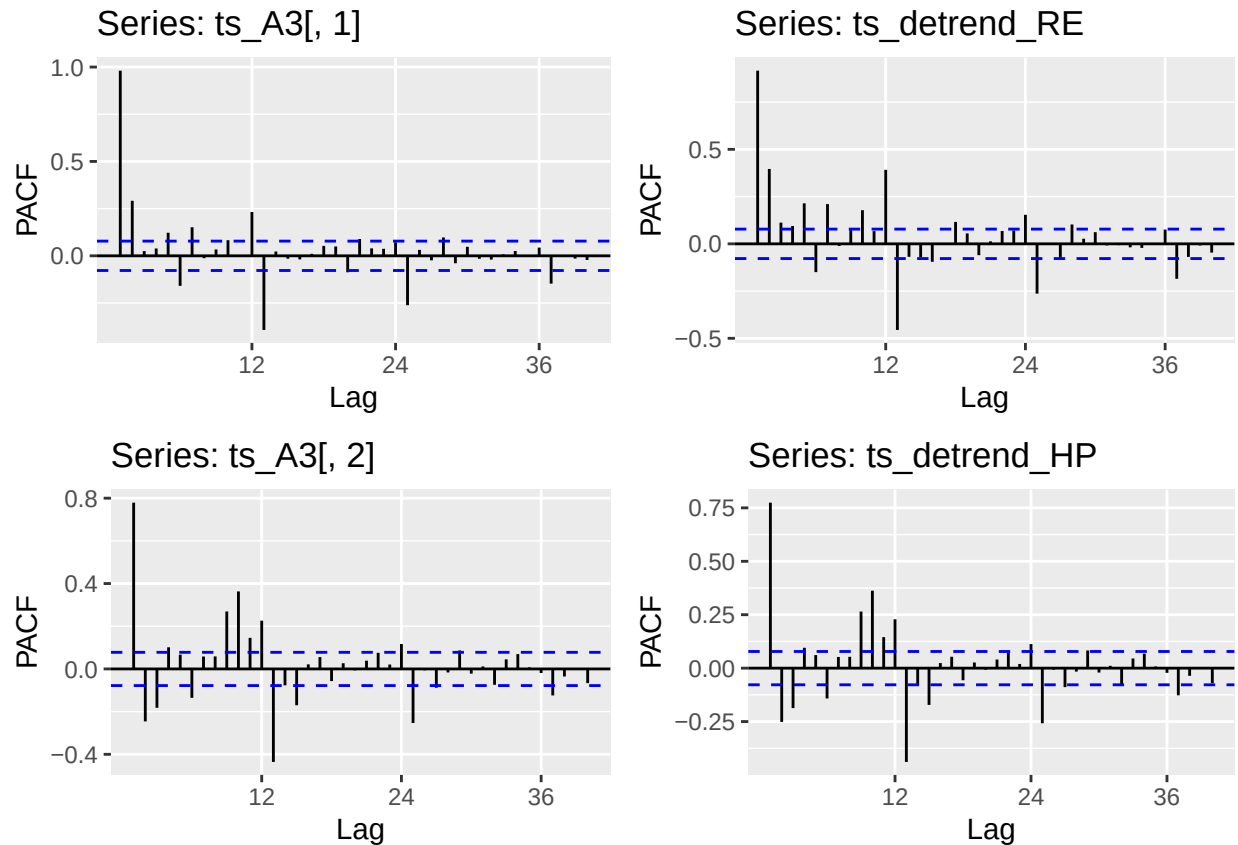
```
#ACF plots
ACF3 <- ggAcf(ts_detrend_RE, lag.max=40, plot=T)
ACF4 <- ggAcf(ts_detrend_HP, lag.max=40, plot=T)

#PACF plots
PACF3 <- ggPacf(ts_detrend_RE, lag.max=40, plot=T)
PACF4 <- ggPacf(ts_detrend_HP, lag.max=40, plot=T)

plot_grid(ACF1, ACF3, ACF2, ACF4)
```

```
plot_grid(PACF1, PACF3, PACF2, PACF4)
```



Because the hydropower timeseries does not have much trend as indicated by the linear regression, I did not expect much change to the ACF - and there is no discernible difference. The renewable did have a trend removed and it slightly decreased the ACF. The PACF also appears unchanged for hydro-power but it does appear changed for renewable energy production. Removing the trend shows possibly more of a seasonal oscillation.

Seasonal Component

Set aside the detrended series and consider the original series again from Q1 to answer Q6 to Q8.

Q6

Just by looking at the time series and the acf plots, do the series seem to have a seasonal trend? No need to run any code to answer your question. Just type in your answer below.

Answer: It appears that the hydropower series has a seasonal trend from both time series and ACF plot. It does not appear renewable has a seasonal trend from the time series plot, but the ACF plot indicates there might be a seasonal aspect (annual spikes down).

Q7

Use function `lm()` to fit a seasonal means model (i.e. using the seasonal dummies) to the two time series. Ask R to print the summary of the regression. Interpret the regression output. From the results, which series have a seasonal trend? Do the results match your answer to Q6?

```
#seasonal means model for renewable energy with the Q1 data, not detrended
RE_dummies <- seasonaldummy(ts_A3[,1])
RE_seasonal_model <- lm(new_df[,2]~RE_dummies)
summary(RE_seasonal_model)
```

```
##
## Call:
## lm(formula = new_df[, 2] ~ RE_dummies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -213.33  -97.36  -59.88  121.55  389.62
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    417.265     21.096   19.779  <2e-16 ***
## RE_dummiesJan     2.090     29.693    0.070   0.944
## RE_dummiesFeb   -34.524     29.693   -1.163   0.245
## RE_dummiesMar     5.956     29.693    0.201   0.841
## RE_dummiesApr    -6.900     29.693   -0.232   0.816
## RE_dummiesMay     8.162     29.693    0.275   0.784
## RE_dummiesJun    -2.231     29.693   -0.075   0.940
## RE_dummiesJul     3.864     29.693    0.130   0.897
## RE_dummiesAug    -3.978     29.693   -0.134   0.893
## RE_dummiesSep   -29.033     29.693   -0.978   0.329
## RE_dummiesOct   -19.937     29.834   -0.668   0.504
## RE_dummiesNov   -20.617     29.834   -0.691   0.490
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 152.1 on 621 degrees of freedom
## Multiple R-squared:  0.008243, Adjusted R-squared: -0.009324
## F-statistic: 0.4692 on 11 and 621 DF, p-value: 0.9223
```

```
#store coeffs
RE_beta0_seasonal <- RE_seasonal_model$coefficients[1]
RE_beta1_seasonal <- RE_seasonal_model$coefficients[2:12]
```

```
#seasonal means model for hydropower with the Q1 data, not detrended
HP_dummies <- seasonaldummy(ts_A3[,2])
HP_seasonal_model <- lm(new_df[,3]~RE_dummies)
summary(HP_seasonal_model)
```

```
##
## Call:
## lm(formula = new_df[, 3] ~ RE_dummies)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -30.895  -6.368  -0.595   6.213  32.557
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    79.724      1.436  55.511 < 2e-16 ***
## RE_dummiesJan   4.951      2.021   2.449 0.014591 *
## RE_dummiesFeb  -2.415      2.021  -1.195 0.232608
## RE_dummiesMar   7.116      2.021   3.520 0.000463 ***
## RE_dummiesApr   5.614      2.021   2.777 0.005649 **
## RE_dummiesMay  14.080      2.021   6.965 8.38e-12 ***
## RE_dummiesJun  10.780      2.021   5.333 1.36e-07 ***
## RE_dummiesJul   4.003      2.021   1.980 0.048091 *
## RE_dummiesAug  -5.320      2.021  -2.632 0.008710 **
## RE_dummiesSep -16.598      2.021  -8.211 1.28e-15 ***
## RE_dummiesOct -16.329      2.031  -8.040 4.56e-15 ***
## RE_dummiesNov -10.782      2.031  -5.308 1.54e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.36 on 621 degrees of freedom
## Multiple R-squared:  0.4714, Adjusted R-squared:  0.4621
## F-statistic: 50.35 on 11 and 621 DF,  p-value: < 2.2e-16
```

```
#store coeffs
HP_beta0_seasonal <- HP_seasonal_model$coefficients[1]
HP_beta1_seasonal <- HP_seasonal_model$coefficients[2:12]
```

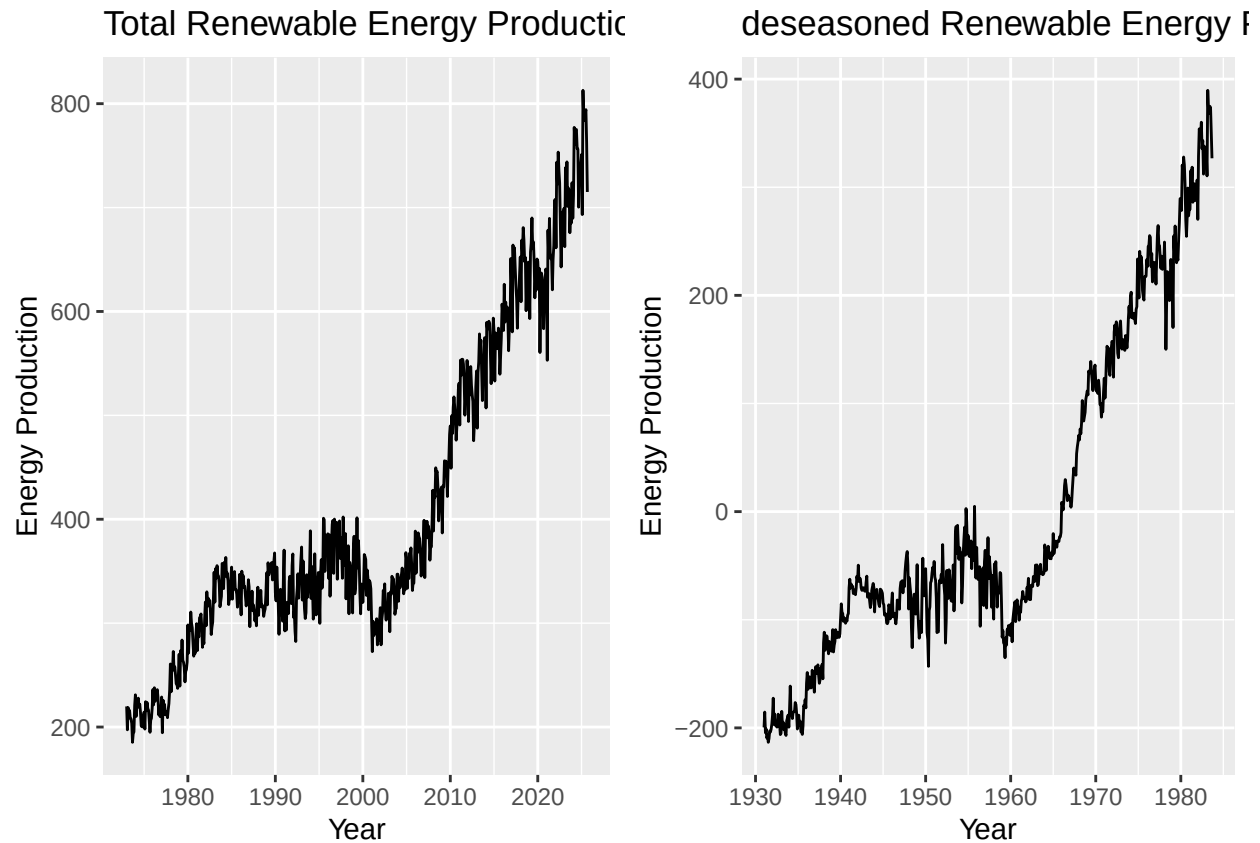
For renewable energy production I have a really low R^2 close to 0 and a terrible p-value. This would indicate no seasonality. My answer for 6 considered it with trending removed so though they dont match, thats ok - might be different if the seasonality was looked for on the detrended time series.

For hydropower consumption, I have a really low p-value (<0.05) and a reasonable R^2 value (0.46), not extremely high but high enough I feel safe saying there is seasonality in the time series.

Q8

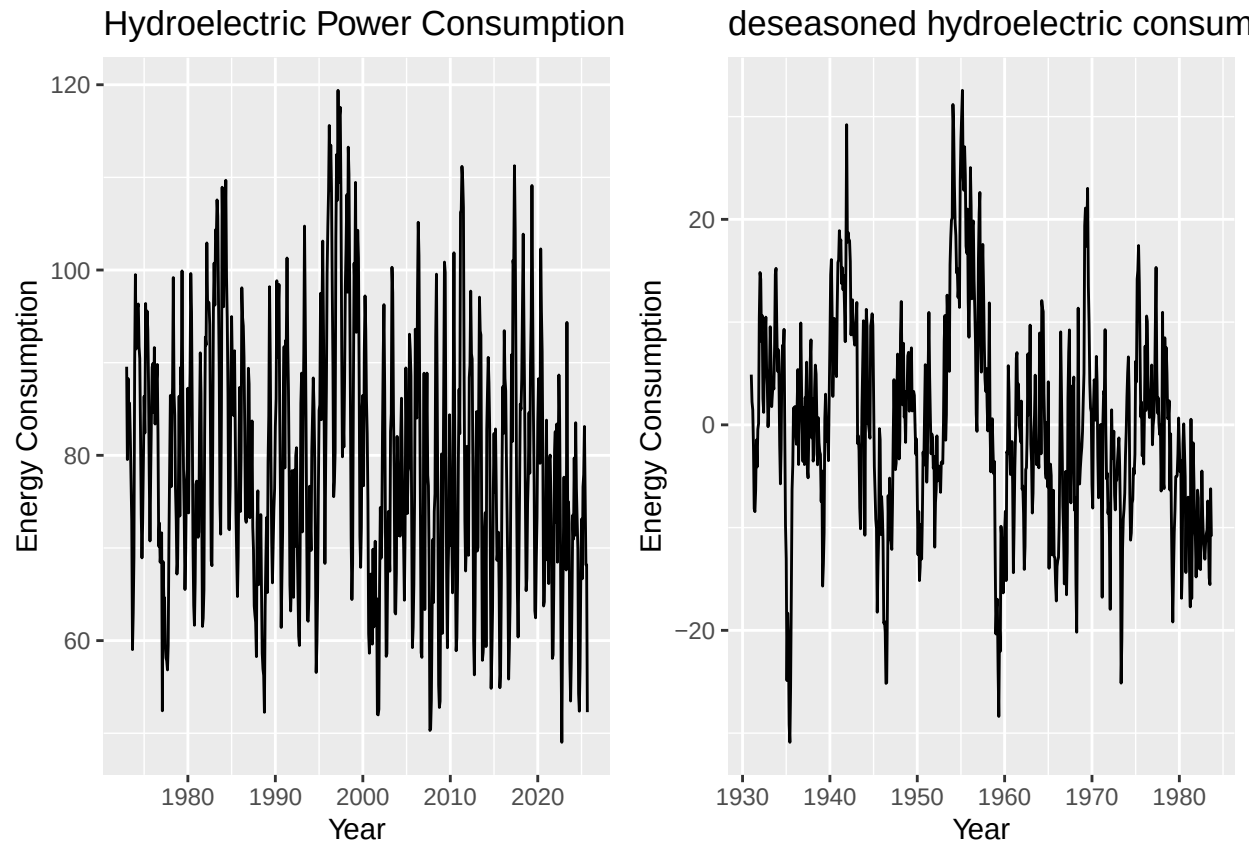
Use the regression coefficients from Q7 to deseason the series. Plot the deseason series and compare with the plots from part Q1. Did anything change?

```
#compute seasonal component for Renewable
season_RE <- array(0, nobs)
for(i in 1:nobs){
  season_RE[i] <- RE_beta0_seasonal+RE_beta1_seasonal**RE_dummies[i,]
#remove from original dataframe for RE
deseason_RE <- new_df[,2] - season_RE
#transform to ts
ts_deseason_RE <- ts(deseason_RE,frequency = 12, start=c(1931,1))
#Plot against RE ts from Q1
ts5 <- autoplot(ts_deseason_RE,
  main = "deseasoned Renewable Energy Production",
  xlab = "Year",
  ylab = "Energy Production")
plot_grid(ts1, ts5)
```



This looks pretty much the same (renewable energy). There was not much seasonality accounted for in the model I made so I'm not too surprised. Some of the fluctuations are now reduced though - I can see minimal changes.

```
#compute seasonal component for hydropower
season_HP <- array(0, nobs)
for(i in 1:nobs){
  season_HP[i] <- HP_beta0_seasonal+HP_beta1_seasonal*HP_dummies[i,]
}
#remove from original dataframe for RE
deseason_HP <- new_df[,3] - season_HP
#transform to ts
ts_deseason_HP <- ts(deseason_HP,frequency = 12, start=c(1931,1))
#Plot against HP ts from Q1
ts6 <- autoplot(ts_deseason_HP,
  main = "deseasoned hydroelectric consumption",
  xlab = "Year",
  ylab = "Energy Consumption")
plot_grid(ts2, ts6)
```



This hydroelectric time series now looks significantly deseasoned. There appears to be something else affecting consumption as it's by no means flat now, but a lot of the smaller scale oscillations are now removed. Perhaps there's a larger scale seasonality as well (scale of years)?

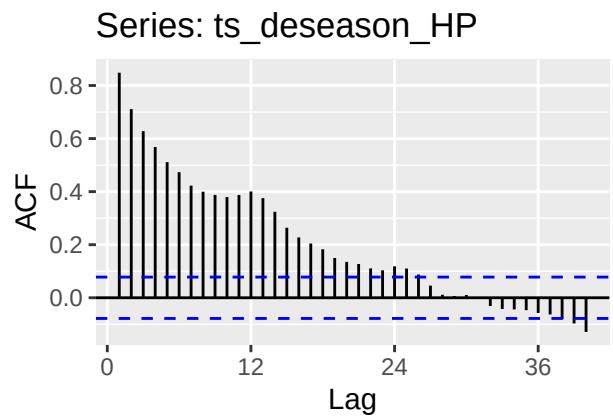
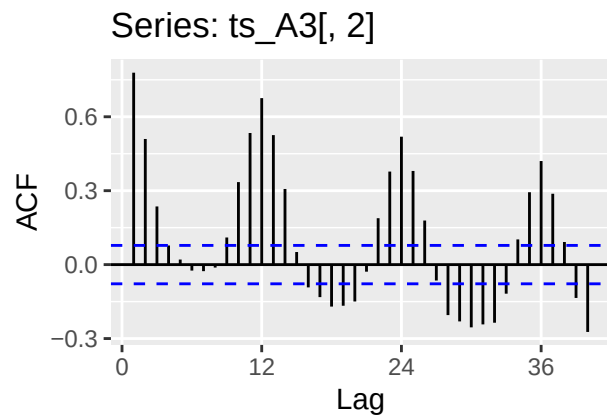
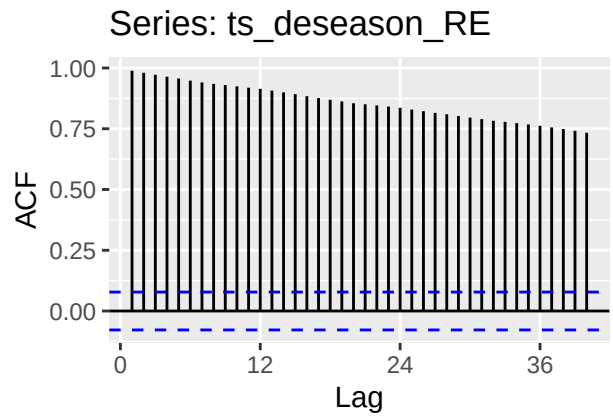
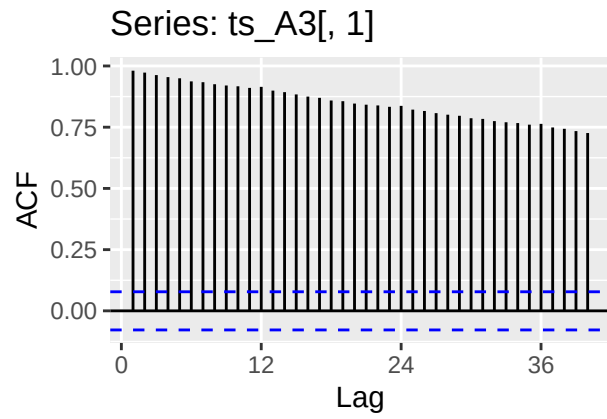
Q9

Plot ACF and PACF for the deseason series and compare with the plots from Q1. You may use `plot_grid()` again to get them side by side. Did the plots change? How?

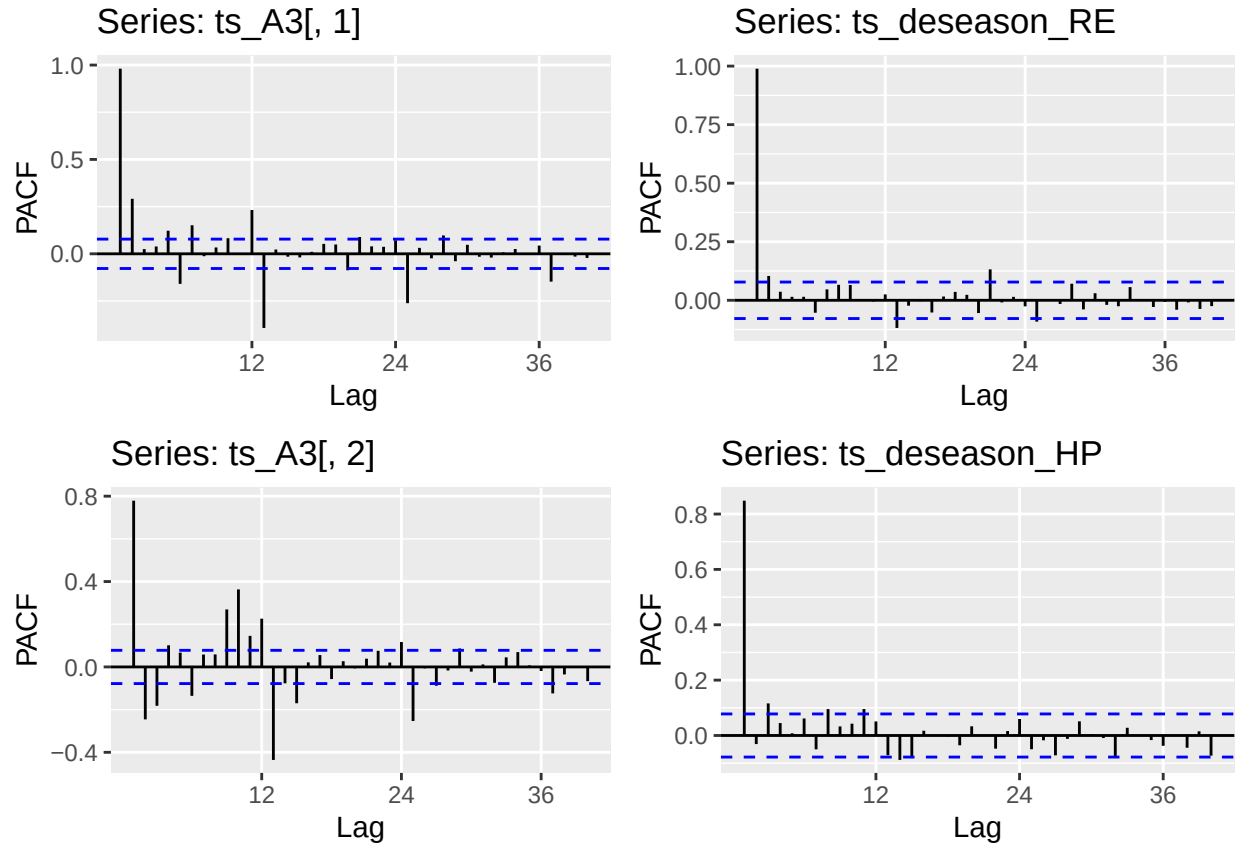
```
#ACF plots
ACF5 <- ggAcf(ts_deseason_RE, lag.max=40, plot=T)
ACF6 <- ggAcf(ts_deseason_HP, lag.max=40, plot=T)

#PACF plots
PACF5 <- ggPacf(ts_deseason_RE, lag.max=40, plot=T)
PACF6 <- ggPacf(ts_deseason_HP, lag.max=40, plot=T)

plot_grid(ACF1, ACF5, ACF2, ACF6)
```



```
plot_grid(PACF1, PACF5, PACF2, PACF6)
```



The ACF for renewable energy production appears completely unchanged. However, the seasonality in the ACF for hydroelectric consumption appears greatly changed. Instead of oscillating the ACF now goes linearly decreasing through the lags. Significant lags end earlier as well.

The PACF for renewable energy production did change, it got rid of most of the seasonally significant lags that popped up once a year in the PACF from Q1. The PACF for hydroelectric consumption shows one significant lag and then almost none after- it removed almost all significant lags which makes sense if all the seasonality is gone.